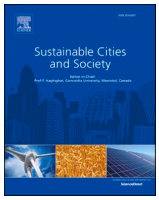
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Assessing and interpreting perceived park accessibility, usability and attractiveness through texts and images from social media

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ABSTRACT

Understanding public perceptions of urban parks is essential for their effective management. While conventional survey methods are resource-intensive, Social Media Data (SMD) offers a cost-effective alternative to gathering public insights. However, using SMD for park perception assessment remains challenges, particularly in integrating text and image analysis and filtering irrelevant content to identify influencing factors across various dimensions. Based on a manually curated dataset, this study introduces the Park Dual-modal Perception (PDP) model, a cutting-edge approach combining SMD text and image analysis to evaluate perceived park accessibility, usability, and attractiveness with an average accuracy of 86.81 %, outperforming the commonly used BERT model by 8.26 %. Utilizing SMD from 130 parks in Guangzhou, the model effectively quantifies the three dimensions, generating visual scoring maps to identify parks with lower perceived scores at the urban scale. Further incorporation of SHapley Additive exPlanations (SHAP) within the PDP model filters 82.79 % of irrelevant words and extracts 158 thematic words and 954 associated words, providing targeted suggestions for park improvements. Our findings indicate that (1) factors such as distance, travel time, ticket prices, and proximity to commercial amenities critically influence park accessibility. (2) Park usability hinges on park's ability to serve diverse groups and provide well-maintained, multifunctional facilities. (3) Park attractiveness is closely linked with the cultural and regulatory characteristics of ecosystem services. Our methodology combines assessment and interpretation of public perceptions at both city and park scales. It aids decision-makers in identifying low-rated parks and understanding the underlying reasons, thereby facilitating more informed urban planning decisions.

1. Introduction

Urban parks are crucial in providing residents with a range of ecosystem services, nature engagement, and recreational opportunities (Wolch et al., 2014), thereby positively impacting their physical (Kaczynski et al., 2008; McCormack et al., 2010), psychological (Cheng et al., 2021; Jiang et al., 2021), and social well-being (Kotabe et al., 2016). The effectiveness of parks in delivering these benefits largely hinges on visitors' actual use, which has been demonstrated to be influenced by the accessibility and quality of parks (Park, 2017; Li & Lin, 2024). Consequently, extensive assessments of these aspects of parks have been widely conducted over the years (Park, 2017; Wang et al., 2015; Wendel et al., 2012; Zhang et al., 2015;). These efforts are crucial for optimizing the distribution and enhancing the quality of parks, aligning with the United Nations' Sustainable Development Goals (SDGs) to promote "good health and well-being (SDG3)" and foster "sustainable cities and communities (SDG11)" (UN, 2015).

Current assessment methods for park accessibility and quality primarily depend on quantifying spatial physical variables. For instance, accessibility is measured using travel distance and time (Bertram & Rehdanz, 2015; Wendel et al., 2011) as well as the density of transportation facilities (Chen et al., 2018; Li et al., 2023). Quality assessments of parks typically consider park characteristics such as recreational facilities (Huai et al., 2023), maintenance (Keunhyun,

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2017; Wang et al., 2015), and aesthetics aspects (Biernacka et al., 2022; Kothencz & Blaschke, 2017; Li & Lin, 2024). However, these objective measurements, such as geographic accessibility, often fail to align with subjective perceptions like perceived accessibility (Jones et al., 2009; Park, 2017). Therefore, there is increasing recognition of the significance of subjective perceptions in understanding and predicting human behaviors, and it's imperative to examine parks from the user's perspective to ensure the urban environment better meets the needs of the public (Biernacka et al., 2022; Byrne & Wolch, 2009).

Public perception of parks represents the common views held by the general public about these spaces, reflecting the emotional interactions between humans and the park environments (Huai & Van de Voorde, 2022; Tang et al., 2023; Zube et al., 1982). Nevertheless, measuring public perception is challenging due to its inherent intangibility and subjectivity (Huai & Van de Voorde, 2022; Tang et al., 2023). Traditional methods mostly rely on cost-intensive manual assessments by experts or respondents analyzing physical scenes, images, or texts. These approaches are laborious, expensive, time-consuming, and can be constrained by limited data sources (Bureau, 2023; Gosal et al., 2019; Liu et al., 2023; Wang et al., 2021), presenting greater challenges at broader spatial and temporal scales.

In the big data era, every citizen acts as a "sensor," significantly increasing the volume of available data (Wang et al., 2021). Social Media Data (SMD) from platforms such as TripAdvisor, Twitter, Yelp, Ctrip, and Dianping provides abundant first-person, multidimensional user-generated content including texts, photographs, and videos (He et al., 2023; Wang et al., 2021). Such data authentically reflects citizens' feelings and experiences of the places they visit (Komossa et al., 2020), showing great potential to overcome traditional limitations and facilitate a comprehensive, in-depth understanding of public preferences and societal insights (Huai & Van de Voorde, 2022; Rui, 2023; Wartmann et al., 2021; Wilkins et al., 2022).

However, analyzing SMD is challenging due to its unstructured and complex nature (Huai et al., 2023; Yan et al., 2024). Current research (Table 1) primarily applies statistical and machine learning-based methods in Natural Language Processing (NLP). Statistics-based methods usually use word frequency analysis to identify key factors influencing perceptions (Huai & Van de Voorde, 2022; Liu et al., 2023; Tang et al., 2023). Another approach involves keyword matching. For instance, Wang et al. (2021) developed an ecosystem services lexicon to assess nine categories of park services by checking if the contents of comments matched words within the lexicon. Moreover, advanced NLP techniques enable the extraction of concepts, entities, and sentiments from texts. Neural networks, in particular, are effective in many NLP tasks (Khurana et al., 2023). Sentiment analysis, which aims to determine the emotional tone behind texts (Wankhade et al., 2022), is relatively mature due to the availability of comprehensive datasets, advanced deep learning algorithms, and open NLP platforms like Google and Baidu. These resources have supported extensive research to convert intangible perceptual information in comments into quantifiable sentiment scores in urban studies (Cheng et al., 2021; He et al., 2023; Huai & Van de Voorde, 2022; Huai et al., 2023; Li et al., 2023; Liu et al., 2023; Rui, 2023; Tang et al., 2023).

However, ready-made sentiment analysis models are typically designed for general scenarios and may not effectively capture specific aspects of human perceptions (Zhou et al., 2024). In response, recent state-of-the-art efforts have focused on developing deep learning models on datasets curated for specific SMD assessment tasks. These include exploring worldwide public sentiments during the COVID-19 pandemic (Rahman et al., 2023), revealing the sentiments and topics of comments toward mobile cabin hospitals (Zhou et al., 2024), and identifying flood locations while extracting water level information in affected areas (Yan et al., 2024). Such targeted approaches not only enhance the applicability of SMD but also enrich our understanding of the environment, thereby increasing its potential benefits for environmental practice (Tang et al., 2023). Nevertheless, datasets specifically curated for park

Table 1
Latest research on public perceptions using SMD.

Reference	Data	SMD assessment methods	Method focus
Wang et al., (2021)	114,000 comments from Dianping about parks in Beijing	Keyword matching (Jieba, HowNet, word2vec), Keyword clustering (K-Means)	Structuring comments to assess nine ecosystem service categories.
Cheng et al. (2021)	350,000 Weibo posts about parks in Nanjing	Sentiment analysis (Baidu NLP)	Evaluating the happiness level of each post.
Huai and Van de Voorde (2022)	21,642 comments from Ctrip and 28,398 comments from Google Maps on parks in Shanghai and Brussels	Sentiment analysis (Baidu, Google NLP), Word frequency analysis (Jieba, TextBlob), Keyword clustering (K-Means)	Quantifying park sentiments and identifying environmental features affecting perceptions.
Huai et al. (2023)	37,657 comments from Dianping and Ctrip about parks in Shanghai	Sentiment analysis (Baidu NLP)	Quantifying park satisfaction scores and comparing them with survey data.
He, Sun et al. (2023)	221,104 Weibo posts from Beijing's metropolitan area	Sentiment analysis (BERT), Topic classification (BERTopic)	Quantifying residential sentiments within urban environments.
Liu, et al. (2023)	5735 comments from Dianping about greenways in Wuhan	Sentiment analysis (Baidu NLP), Word frequency analysis (Jieba)	Examining sentiment preferences towards greenways.
Li et al. (2023)	111,814 comments from Dianping and Weibo about parks in Shanghai	Sentiment analysis (LSTM)	Quantifying public satisfaction with parks.
Tang et al. (2023)	152,072 comments from Ctrip, Tongcheng, and Qyer about attractions in Beijing and Shenzhen	Sentiment analysis (Tencent NLP), Word frequency analysis (Jieba)	Understanding public likes and dislikes regarding urban attractions and their environmental characteristics.
Rui (2023)	135,650 Weibo posts from urban villages and formal settlements in Shenzhen	Sentiment analysis (BERT trained on an open dataset)	Measuring positive sentiments among residents in urban villages and formal settlements.
Rahman et al. (2023)*	12,000 coronavirus-related Twitter posts from the USA, UK, and Bangladesh	Sentiment analysis (4 machine learning models trained on a manually annotated dataset), Word frequency analysis	Comparing sentiment models, analyzing global sentiment variations, and identifying key drivers.
Zhou, Wang et al. (2024)*	413,572 Weibo posts on Mobile Cabin Hospitals (MCH) in China	Sentiment analysis, Topic classification (BERT-based models trained on annotated posts)	Classifying MCH-related comments into three sentiments and four primary topics.
Yan et al. (2024)*	75,868 flood-related Weibo posts from Anhui	Address extraction (BERT-BiLSTM-CRF trained on a flood location corpus), Multimodal water depth extraction (BERT-BiLSTM-CRF, VGG16)	Identifying flood locations and extracting water levels.
This study*	62,106 texts and 111,488 images from Ctrip and Dianping about parks in Guangzhou	Text classification (Park Dual-modal Perception model trained on a manually annotated dataset), SHAP analysis	Assessing park comments across three perception dimensions, extracting thematic words and associated words for detailed interpretation.

Note: Studies marked with * developed specific datasets and trained deep learning models for targeted tasks.

perception assessment tasks are limited, and research regarding perceptions beyond sentiment remains underexplored, posing challenges in extending assessments to more focused dimensions.

Apart from mainstream text analysis in SMD, images also reflect public preferences. The application of computer vision for automated classification and recognition of landscape features in SMD images has proven highly effective in quantifying preferences for recreational cultural ecosystem services (Gosal et al., 2019; Runge et al., 2020; Wilkins et al., 2022). Additionally, people tend to post images and texts together on social media to express their emotions or opinions (Zhu et al., 2022), but existing research on park perceptions typically focuses on single-modal analyses, either texts or images. Notably, Yan et al. (2024) integrated text and image data from social media to analyze flooding depth, while Chen et al. (2021, 2023) combined satellite images with crowd-sourced geospatial data to map urban green spaces. These approaches demonstrate the effectiveness of multi-modal approaches, which further highlights the urgent need for a comprehensive methodology that simultaneously analyzes texts and images in SMD to fully assess park perceptions.

Finally, subsequent analyses of perception assessments often involve extracting high-frequency keywords, which are then categorized through word embedding models to examine how environmental features influence perceptions (Huai & Van de Voorde, 2022; Wang et al., 2021; Wartmann et al., 2021). However, this interpretation method, relying mainly on word frequency statistics following text segmentation, tends to overlook contexts, word meanings, grammatical structures, and word relationships. The presence of many irrelevant words can complicate the extraction process, necessitating manual intervention. When evaluating public attitudes, such as accessibility from comments, humans typically make judgments by checking statements related to “transportation”. Similarly, deep learning models that accurately predict perceptions from texts need to understand relationships between words and should have targeted focuses on specific words during analysis. SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) is an explainable algorithm designed to clarify the opaque mechanisms behind artificial intelligence models and reveal their decision-making focuses. In its official example (https://github.com/shap/shap), SHAP identifies "great movie" as a key phrase influencing positive sentiment in the sentence "What a great movie! …if you have no taste," aligning closely with human analytical processes. Yet, the potential for SHAP to enhance the interpretation of perception assessment results still requires further exploration.

Addressing existing research gaps, this study aims to explore the potential of integrating SMD texts and images to establish a comprehensive workflow that synergizes both the assessment and interpretation of park perceptions. We selected perceived accessibility and two typical quality attributes, usability, and attractiveness, as key perception indicators. Parks must be accessible to be usable, and those that are both accessible and usable are often perceived as more attractive (Abdelhamid & Elfakharany, 2020; Iwarsson & Ståhl, 2003; Li & Lin, 2024; Park, 2017; Viinikka et al., 2023; Wendel et al., 2012; Zhang et al., 2015). Our proposed workflow includes (i) SMD acquisition, (ii) establishment of an annotated park perception dataset, (iii) training and evaluation of the Park Dual-modal Perception (PDP) model, (iv) assessment of park perceptions using the PDP model, and (v) interpretation of these perceptions by combining the PDP model with SHAP. To demonstrate the effectiveness of our workflow, we analyzed 62,106 comments from 130 parks in Guangzhou. The perception score maps generated from the assessment session help planners identify low-rated parks on the urban scale, while thematic words and associated words extracted from the interpretation session offer park-level enhancement suggestions, facilitating more informed urban planning decisions. Overall, this study seeks to answer the following questions:

(1) How can texts and images from SMD be used concurrently to accurately assess perceived park accessibility, attractiveness, and usability?
(2) How to efficiently filter irrelevant content to precisely extract keywords and associated words from SMD across the three dimensions, thereby interpreting the assessment results?
(3) What factors influence people’s perceptions of park accessibility, usability, and attractiveness?

2. Materials and methods

2.1. Key concepts

Accessibility refers to the ease with which parks can be reached from one location, providing a measure that evaluates the relative opportunity for contact or use (Gregory et al., 2011; Park 2017). Usability is often used in parallel with accessibility and is defined by attributes that make parks fit for use, such as being functioning, operational, serviceable, valid, and working. It concerns the fulfillment of functional requirements within parks (Iwarsson & Ståhl, 2003; Li & Lin, 2024). Attractiveness relates to how well parks meet users’ personal needs, expectations, and preferences, influencing individuals’ decisions to use and spend time in these spaces (Biernacka et al., 2022; Li & Lin, 2024).

2.2. Research framework

The research framework is illustrated in Fig. 1. We collected 62,106 comments from two popular social media platforms in China and selected 3140 comments to create an annotated dataset in terms of accessibility, usability, and attractiveness. This dataset was used to train the Park Dual-modal Perception (PDP) model, which then served to assess and interpret perceptions reflected in all 62,106 comments. In the assessment section, the model quantified perception scores across the three dimensions, and visual scoring maps generated by ArcMap helped to identify low-rated parks at the macro scale. Further, by integrating SHAP within the PDP model, we were able to filter out irrelevant content and precisely identify factors contributing to score variations, providing strategic suggestions for micro-scale enhancements.

2.3. Study area

This study selects Guangzhou city in China as the case study area. Known for its high-density and compact urban pattern, Guangzhou has experienced significant growth and development. From 2000 to 2022, the city’s population increased from 7.13 million to 10.35 million, while the built-up area expanded from 285 square kilometers to 1380 square kilometers. Simultaneously, Guangzhou has prioritized the development of its green space system, improving the greening coverage rate in built-up areas from 29.34 % to 44.20 %, and increasing the number of parks from 68 to 387 (Bureau, 2023). The ’Guangzhou Green Space System Planning (2021—2035)’ aims to further enhance environmental quality by increasing the per capita park area from 0.61 per 10,000 people to 0.90 per 10,000 people by 2035 and expanding the service radius coverage of green spaces from 80 % to 90 %. These initiatives position Guangzhou as a prime example for park research in rapidly developing urban contexts.

2.4. Data acquisition

We selected Ctrip (https://www.ctrip.com) and Dianping (https://www.dianping.com), two of China’s most popular social media platforms, as our primary data sources. These platforms are widely recognized in park research for enabling users to freely share their experiences and comments about parks (Huai et al., 2023; Huai & Van de Voorde, 2022; Li et al., 2023; Tang et al., 2023; Wang et al., 2021). Our data collection covered 387 established parks in Guangzhou, with

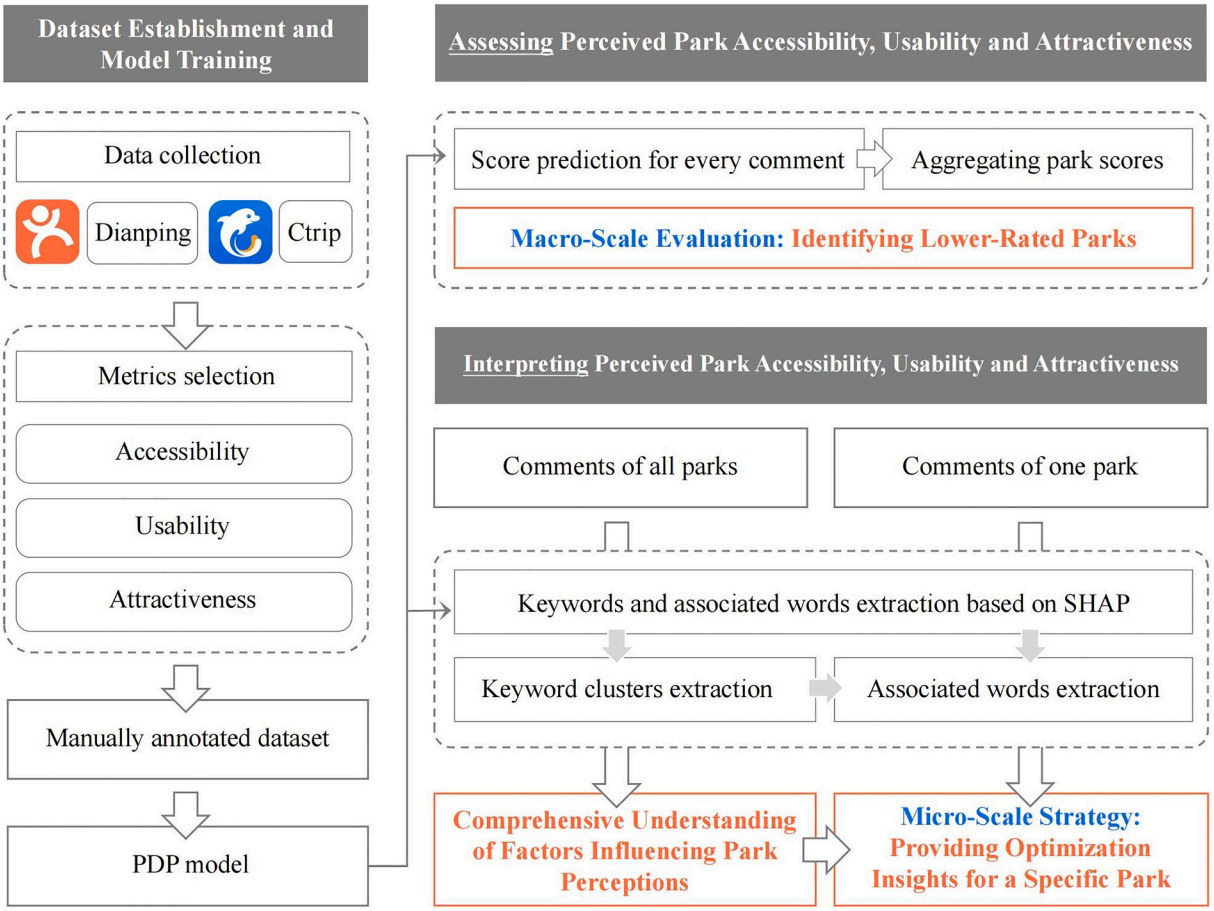


Fig. 1. Research framework.

Python scripts used to access the APIs of these platforms. We collected all comments available up to August 1, 2023. To guarantee adequate data samples for each park, we included only those with over 30 combined comments on both platforms (Huai & Van de Voorde, 2022), a criterion met by 130 parks (Fig. 2). Following data cleaning to exclude blank, irrelevant, and duplicate comments, we compiled 62,106 valid texts and 111,488 images. Each comment record includes the username, timestamp, comment texts, and any associated images.

2.5. Manually annotated dataset

From the collected data, we randomly selected 2500 comments, which, along with their texts and images, were evaluated on accessibility, usability, and attractiveness. Each comment was labeled as '0' (negative), '1' (positive), or '2' (not relevant) across three dimensions. To ensure data balance and optimize our deep-learning model's performance, we expanded and carefully curated the selection of comments based on these labels. The final dataset, consisting of 3140 comments, was collaboratively annotated by two experts. In cases of annotation discrepancies, thorough discussions were conducted to reach an agreement, thus ensuring consistent and scientifically valid data for model training.

2.6. Development of the park dual-modal perception (PDP) model

The development of the PDP model (Fig. 3) is inspired by the latest advancements of OpenAI's Contrastive Language-Image Pre-Training (CLIP) model (Kaplan et al., 2020). CLIP employs contrastive learning on an extensive corpus of 400 million internet-derived image-text pairs. This model effectively aligns text with corresponding image features and demonstrates exceptional adaptability, achieving state-of-the-art performance in zero-shot image classification tasks across various datasets.

Our PDP model builds upon the CN-CLIP framework (Yang et al., 2022), enabling effective extraction of information from both the textual and image data. We fine-tuned this model using the dataset developed in Section 2.4 (manually annotated dataset). Loss is calculated using a contrastive loss function designed to minimize the distance between positive pairs and maximize it for negative pairs. The function's formula is:

$$\text{Loss} = -\sum_{i=1}^{N} \log \left(\frac{\exp(\langle f_i^I, f_i^T \rangle / \tau)}{\sum_{j=1}^{N} \exp\left(\langle f_i^I, f_j^T \rangle / \tau \right)} \right) \quad (1)$$

Here, $\langle f_i^I, f_j^T \rangle$ denotes the dot product between the normalized feature vector of the *i*-th image and *j*-th text, serving as a measure of similarity, and τ represents a temperature scaling parameter.

The fine-tuned model, named Park-CN-CLIP, includes a text encoder (RoBERTa-wwm-ext-base-chinese) and an image encoder (ViT-B-16). The subsequent training steps include:

(1) **Feature Extraction:** Utilizing the text and image encoders of Park-CN-CLIP to extract features T_i and I_i from the texts and images.
(2) **Feature Fusion:** Merging these features using a transformer encoder layer that employs a multi-head self-attention mechanism, expressed as:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^O \quad (2)$$

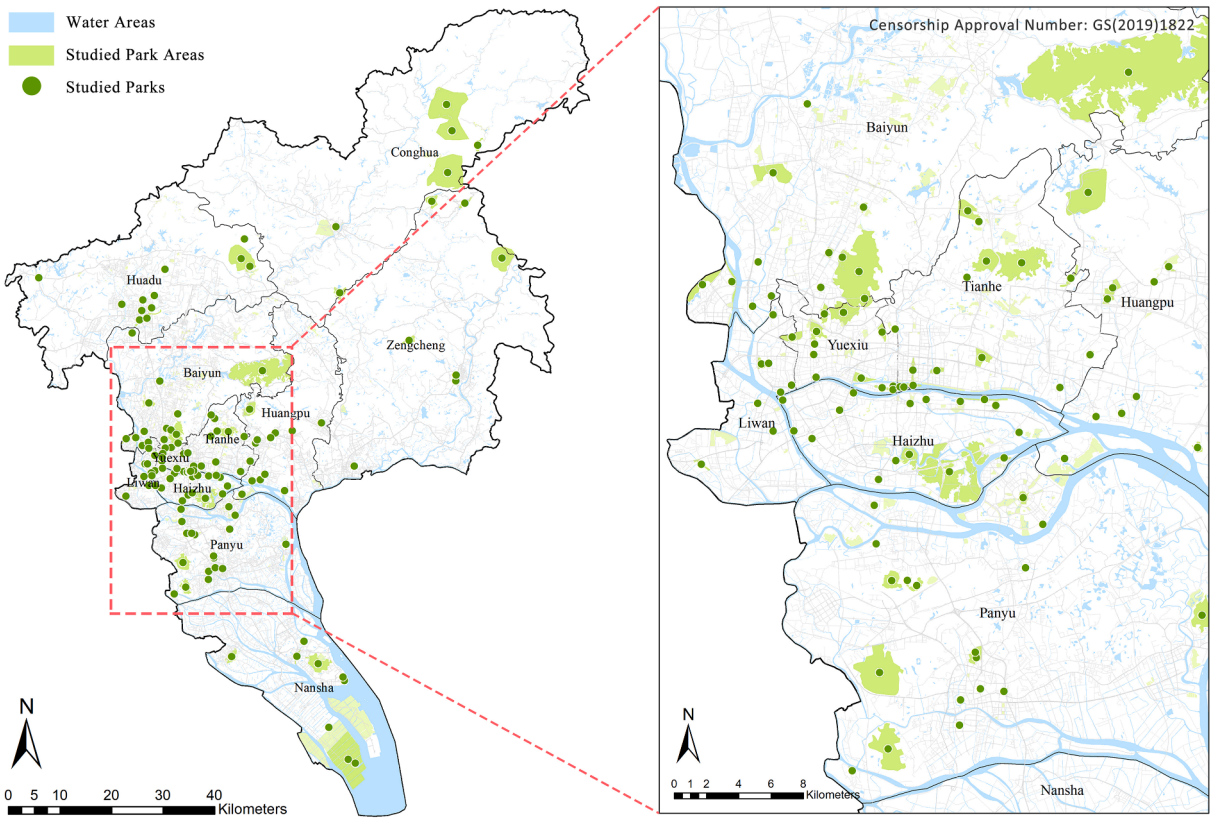


Fig. 2. Study area and geographical spatial distribution of the studied parks.

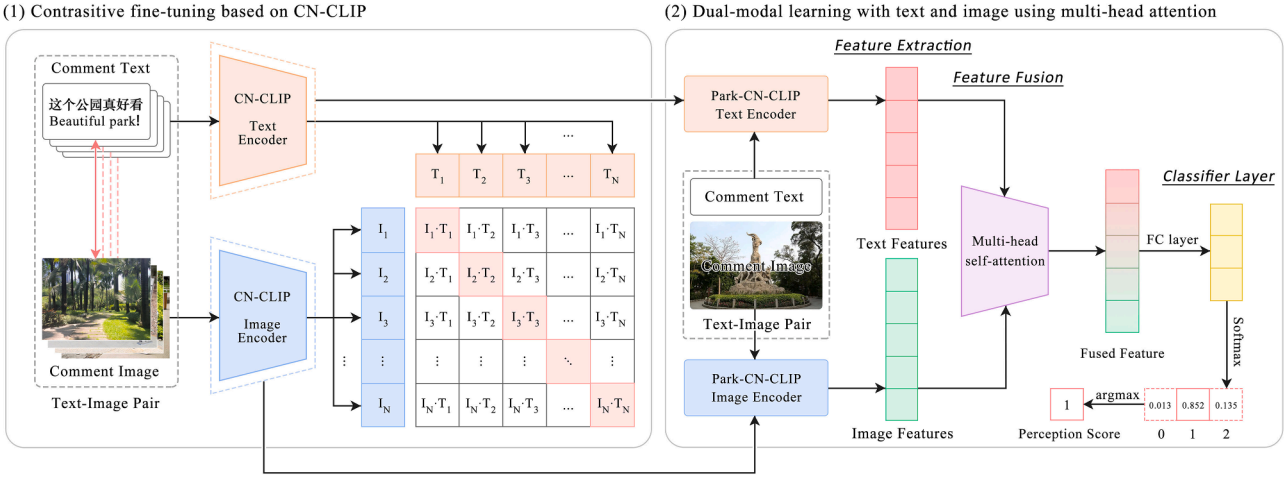


Fig. 3. Structure of the PDP model.

Where each $head_i$ is computed as:

$$head_i = Attention(QW_i^Q,\ KW_i^K,\ VW_i^V) \tag{3}$$

The Attention function uses scaled dot-product attention:

$$Attention(Q,\ K,\ V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

Here, W_i^Q, W_i^K, W_i^V, and W^O are parameter matrices. d_k is the dimensionality of the key vectors, Q, K, Vare query, key, and value vectors of the merged feature $[T_i : I_i]$, respectively.

(3) **Classifier Layer:** Implementing a Fully Connected (FC) layer as a classifier for output prediction. It incorporates a ReLu activation function and Dropout regularization to prevent over-fitting. The formulation of the FC classifier function is:

$$z = Wx + b \tag{5}$$

Where x is the input to the classifier, *W* represents the weight matrix of the layer, b is the bias vector, and z is the output of the linear transformation before activation.

(4) **Performance Evaluation:** The model employs the CrossEntropyLoss function during training to optimize parameters, followed by the backpropagation algorithm to update weights effectively. The CrossEntropyLoss function is detailed below:

$$H(y,\ \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{10} y_{ij}\log\widehat{y_{ij}} \tag{6}$$

where y_{ij} is the label indicating whether the i th sample belongs to class c, and $\widehat{y_{ij}}$ is the probability predicted by the model for the i th sample belonging to class j, with N being the number of samples.

We use accuracy (ACC) as the metric to evaluate the model's effectiveness, calculated as follows:

$$ACC = \frac{N_{\hat{y}_i=y_i}}{n} \tag{7}$$

where *n* represents the total number of comments. For each comment, y_i denotes the actual label, while $\hat{y}_i$ is the model's predicted label for the i th comment. $N_{\hat{y}_i=y_i}$ signifies the number of comments that are correctly classified.

2.7. Assessing and interpreting perceived park accessibility, usability, and attractiveness

2.7.1. Assessment based on the PDP model

In the assessment section, we utilized the fine-tuned PDP model to assign a label of '0′, '1′, or '2′ to each comment from 130 parks. We then calculated an aggregated score for each park by dividing the number of comments labeled as "1" by the total number of comments labeled as "1" and "0". Notably, some comments contained only texts without accompanying images. To maintain consistency in model inputs, a solid black RGB image was used as a placeholder for such text-only comments, allowing the model to rely solely on text features for prediction.

2.7.2. Interpretation based on the PDP model and SHAP

Each comment may contain multiple pieces of information, and relying solely on word frequency for keyword extraction could lead to misleading results due to the presence of irrelevant content. To address this, we used SHAP values (Lundberg & Lee, 2017) to highlight how the PDP model focuses on each character within the text, aiding in the extraction of dimension-specific phrases, keywords, and associated words. The methodology is shown in Fig. 4. and involves the following steps:

(1) **Comment Classification:** We employed the PDP model to select comments labeled as "0" or "1" across the three dimensions. Each selected comment underwent the subsequent analyses.
(2) **SHAP Value Calculation:** Employing the PDP model and checkpoint weights from Section 2.6 (development of the PDP model), we defined a SHAP interpreter to calculate SHAP values for every character in the comments. The formula for SHAP values is as follows:

$$\varnothing_j = \sum_{S\subseteq F\setminus\{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!}\left[f_S\left(x_{S\cup\{j\}}\right) - f_S(x_S)\right] \tag{8}$$

Where $\varnothing_j$ is the SHAP value for feature j, F is the set of all features, S is a subset of features excluding feature j, $|S|$ is the number of features in subset S, $|F|$ is the total number of features, $f_S\left(x_{S\cup\{j\}}\right)$ is the prediction of the model using the features in subset S along with feature j, $f_S(x_S)$ is the prediction of the model using only the features in subset S.

(3) **Key Phrase Extraction:** We summed the SHAP values of all characters in a phrase to calculate its total SHAP value. The top five phrases with the highest SHAP values from each comment were selected for further analysis.
(4) **Keyword Vectorization:** Given that Chinese vocabulary often contains multiple characters (for example, the word for "park" in Chinese, "公园", is composed of two characters: "公" and "园"), we used Jieba for segmentation and the HIT stop-word lexicon to remove redundant words. From the compiled phrase lists, we extracted the top 50 high-frequency keywords and converted them into 100-dimensional word vectors using the Tencent AI Lab Term Embedding Corpora (Song et al., 2018). This corpus includes 8 million Chinese vocabulary and adeptly captures the semantic nuances of Chinese words and phrases.
(5) **Keyword Clustering:** We employed the K-Means algorithm to cluster the high-frequency keyword vectors and then applied the UMAP algorithm to reduce their dimensionality to two dimensions for visualization. The spatial distribution of clusters was presented through a scatter plot, where the size of each point reflects the word frequency and colors differentiate the clusters.
(6) **Associated Word Extraction:** We summarized the associated words for each cluster based on the thematic words and the phrases from the "Text Segmentation and Phrase Formation" procedure.

3. Results

3.1. Training and evaluating the PDP model

3.1.1. Experiments settings

In the training section, we divided the dataset into a training set and a validation set at an 8:2 ratio, with the data distribution presented in Table 2. To avoid over-fitting, considering the size of the dataset, we set the batch size to 32 and the total number of training epochs to 20. For image processing, we employed image augmentation strategies such as random cropping, flipping, and color distortion to enhance the model's generalization ability. The AdamW optimizer was used for updating model parameters.

To validate the effectiveness of the PDP model, we conducted a series of comparative experiments against established models in the urban perception assessment studies. BERT, renowned for its deep bidirectional architecture and rooted in Transformer architecture, has been applied to sentiment analysis of the built environment by Rui (2023) and He et al. (2023). LSTM, a Recurrent Neural Network (RNN) architecture designed to process sequential data and manage long-term dependencies, has been used in park satisfaction analysis by Li et al. (2023). Furthermore, XGBoost, used by Wang et al. (2021) and Rahman et al. (2023) for analyzing public sentiments, is an optimized gradient boosting library widely applied in various classification and regression tasks.

3.1.2. Experimental results

The experimental results presented in Table 3 and illustrated by the training curve in Fig. 5 showcase the effectiveness of our PDP model. This model achieved high prediction accuracies of 0.8628 for accessibility, 0.8489 for usability, and 0.8926 for attractiveness. These figures significantly exceed those achieved by solely using the Park-CN-CLIP's text or image encoders and outperform the widely-used BERT model by 3.17 %, 7.19 %, and 14.43 %, respectively. Impressively, the Park-CN-CLIP Text Encoder alone performs better than the BERT model, with improvements of 0.98 %, 4.41 %, and 11.85 % across the three metrics.

In comparison, traditional machine learning models such as LSTM and XGBoost achieved lower prediction accuracies, indicating their limited effectiveness in accurately predicting human perceptions. Overall, the superior performance of our PDP model highlights its robust capability in reliably assessing human perceptions through SMD.

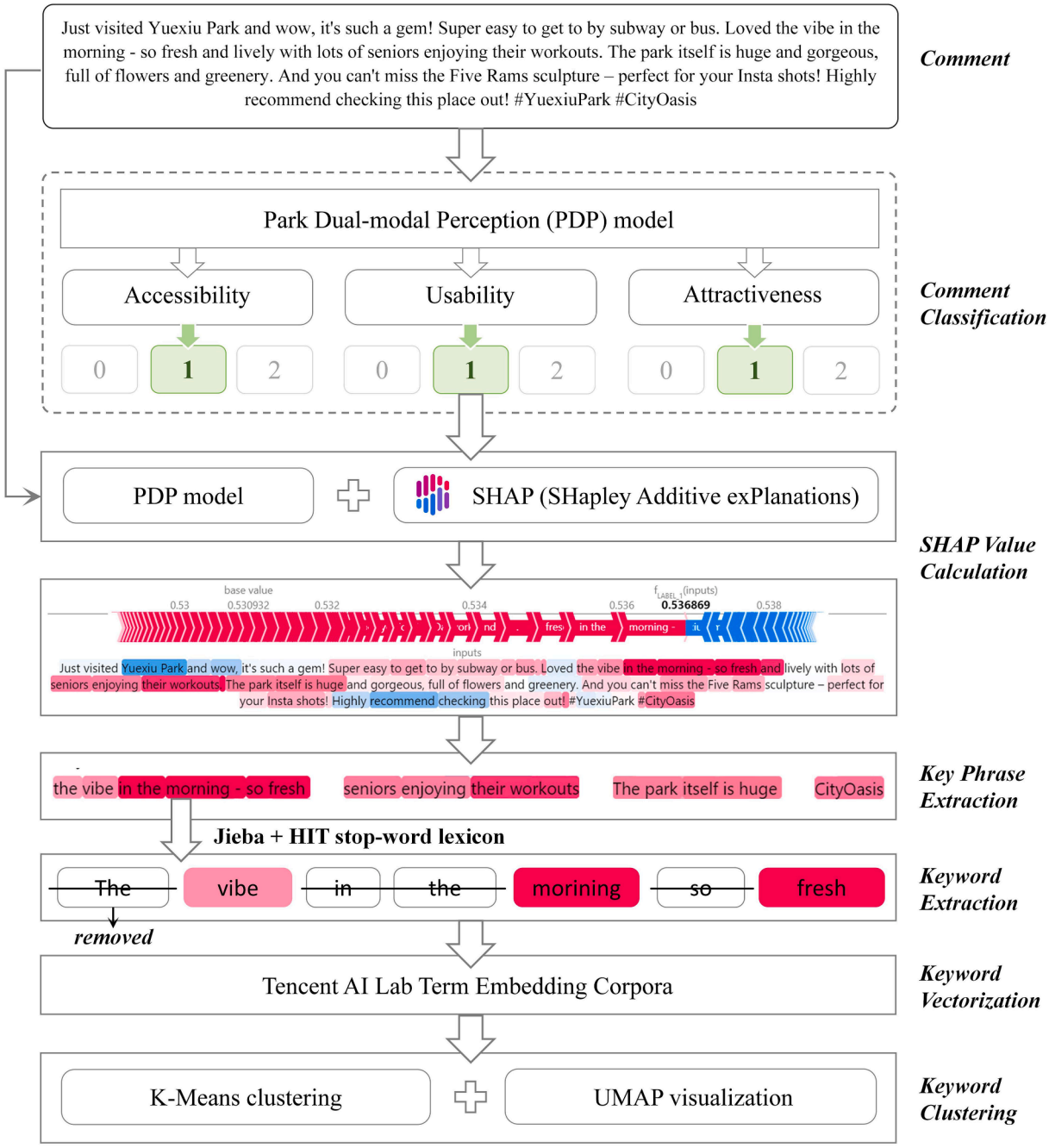


Fig. 4. Interpretation procedure of integrating SHAP within the PDP model. Usability is presented as an example to show the steps after "SHAP Value Calculation". The same procedure applies to the other two dimensions.

Table 2
Overview of data distribution.

Class	Accessibility	Usability	Attractiveness
0	508	336	614
1	937	1523	1816
2	1695	1281	710

Table 3
ACC of different models.

Model	Accessibility	Usability	Attractiveness
PDP model (ours)	**0.8628**	**0.8489**	**0.8926**
Park-CN-CLIP Text Encoder (ours)	0.8409	0.8211	0.8668
BERT	0.8311	0.7770	0.7483
Park-CN-CLIP Image Encoder (ours)	0.5427	0.6024	0.5527
LSTM	0.5291	0.5054	0.6023
XGBoost	0.4915	0.4385	0.5017

Note: All models listed here are trained on our park perception dataset.

3.2. The assessment results within the case study area

Each of the 62,106 comments obtained was assessed from the three dimensions (Table 4). Among these, 13,329 comments (21.5 %) were not labeled as "2" (not relevant), indicating relevance to perceptions of accessibility. Specifically, 3667 comments were labeled as "0"

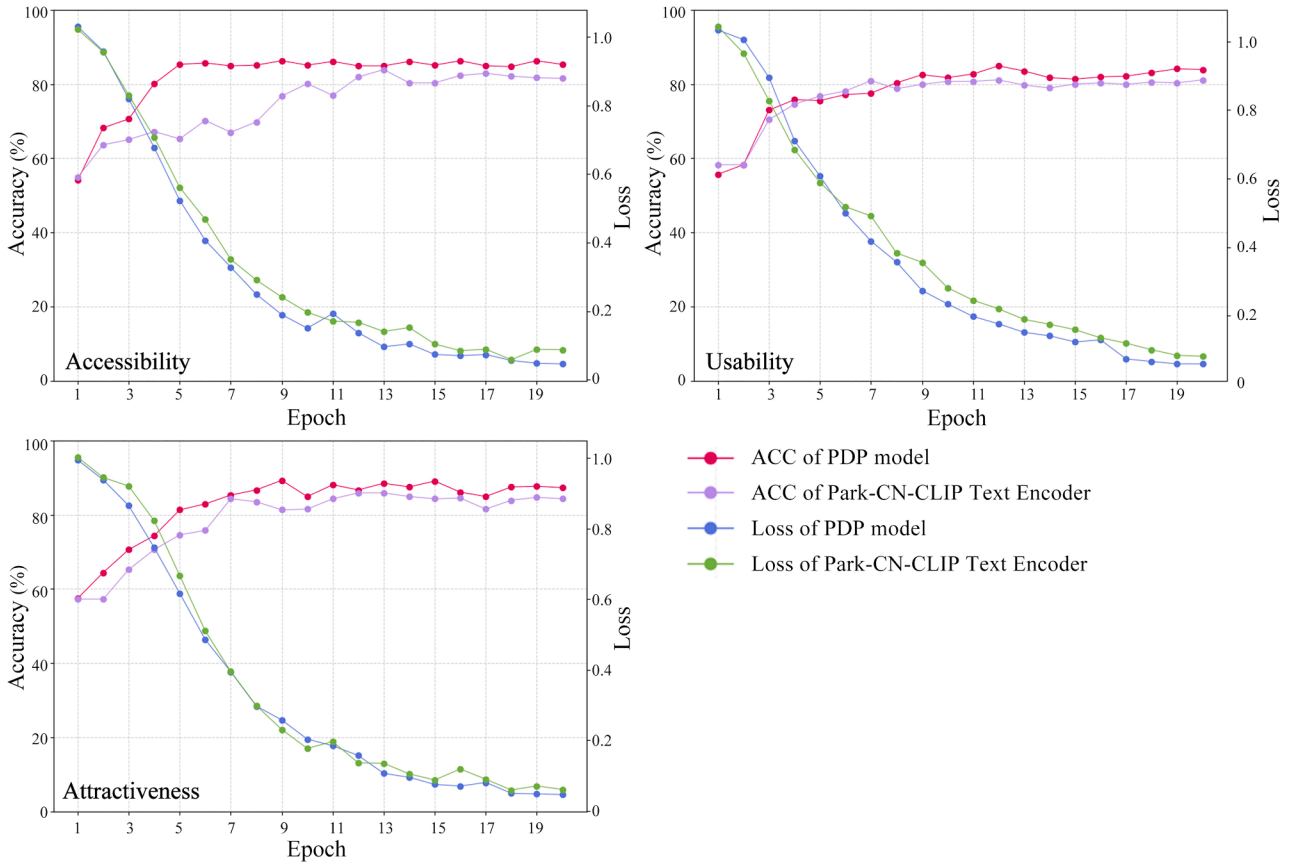


Fig. 5. Training curves of the accuracy and loss on the test set regarding 2 models.

Table 4
Descriptive analysis of score distribution.

	Accessibility		Usability		Attractiveness	
Scores	Number	Percentage	Number	Percentage	Number	Percentage
0.0–0.2	3398	0.255	2253	0.093	5323	0.115
0.2–0.4	247	0.019	111	0.005	302	0.007
0.4–0.6	1399	0.105	3656	0.151	6384	0.138
0.6–0.8	2424	0.182	6918	0.286	13,974	0.303
0.8–1.0	5861	0.440	11,263	0.465	20,137	0.437
In total	13,329	1.000	24,201	1.000	46,120	1.000

(negative), and 9662 as "1" (positive), suggesting that 72.49 % of the feedback positively reflected park accessibility. However, the high variance (0.0361) and a considerable proportion of negative comments (27.51 %) on accessibility indicate varied accessibility scores across parks.

Regarding usability, approximately 39 % of the comments (24,201) addressed this aspect. Among these, 2342 were negative and 21,859 were positive, indicating that 90.32 % expressed positive views on park usability. Attractiveness was the most frequently mentioned dimension (74.3 %, 46,123 comments), highlighting its significance to the public. Of these comments, 5574 were labeled as "0" and 40,546 as "1," showing that 87.90 % positively rated park attractiveness. Both usability and attractiveness achieved high average scores of 0.892 and 0.865, respectively, with variances below 0.01, reflecting consistently high ratings across most parks.

We visualized the score distribution for each dimension using ArcMap 10.6 (Fig. 6). Spatially, parks in Yuexiu and Liwan districts exhibited high levels of accessibility, usability, and attractiveness (average values: 0.8796, 0.9331, and 0.9251, respectively). Interesting contrasts were observed in nearby districts like Panyu, Haizhu, Baiyun, and Tianhe, where Panyu (0.7660, rank 5), Haizhu (0.7942, rank 4), and Baiyun (0.7656, rank 6) had lower accessibility but higher usability (0.8959, 0.9026, 0.9390) and attractiveness (0.8916, 0.8542, 0.9079). Tianhe, on the other hand, showed good accessibility (0.8010, rank 3) but lower usability (0.8408, rank 9) and attractiveness (0.8012, rank 9). In districts like Huangpu, Zengcheng, Conghua, Huadu, and Nansha, accessibility scores decreased progressively, but usability and attractiveness did not show significant differences.

The regression analysis revealed strong positive correlations among the three dimensions. Overall, a consistent pattern emerged in the score distribution across dimensions, with central urban parks receiving higher public perception scores than those in suburban areas. This alignment with the actual characteristics of the administrative areas confirms the effectiveness of our methodology in assessing park perceptions.

3.3. The interpretation results

3.3.1. Keyword extraction results based on SHAP

Although comments often contain multiple layers of information, our

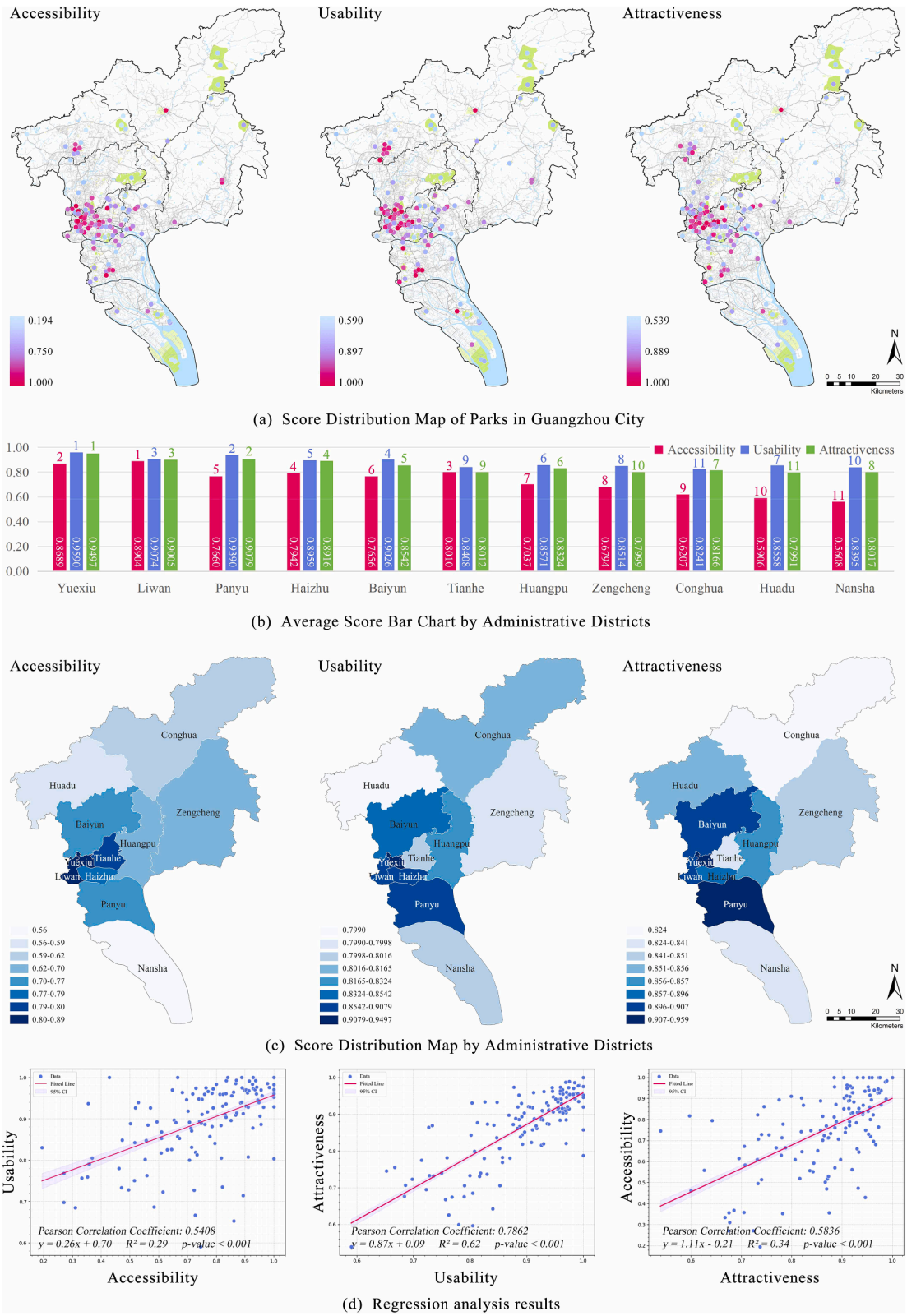


Fig. 6. The measurement results within the case study area. (a) Aggregating score for every park. (b, c) Average scores by administrative district. (d) Regression analysis results.

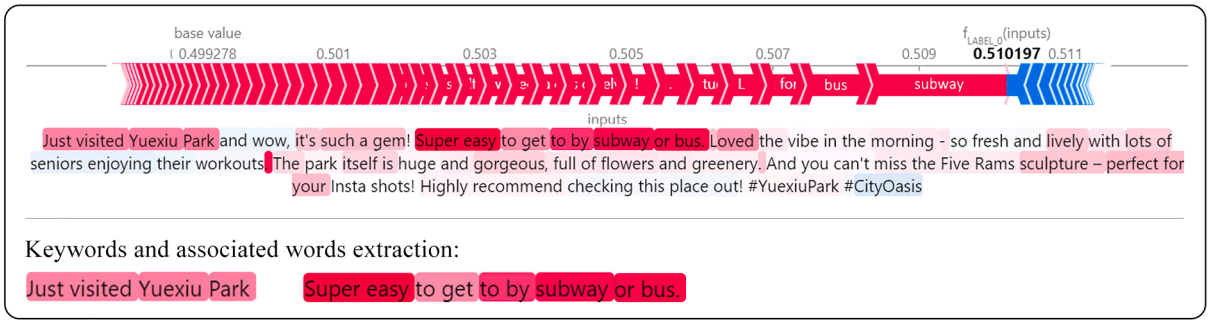
SHAP-based method effectively avoids the disruption caused by irrelevant information and accurately extracts dimension-specific information. For instance, when assessing accessibility, the model focuses on phrases like “super easy to get to by subway or bus” in the example comment, while assigning lower SHAP values to characters irrelevant to accessibility, thereby reducing their emphasis (Fig. 7).

In comparison, we processed all comments through direct word segmentation, removing stop words and punctuation, followed by a token count summary (Table 5). Results show that our SHAP-based method effectively filtered out numerous irrelevant words, capturing

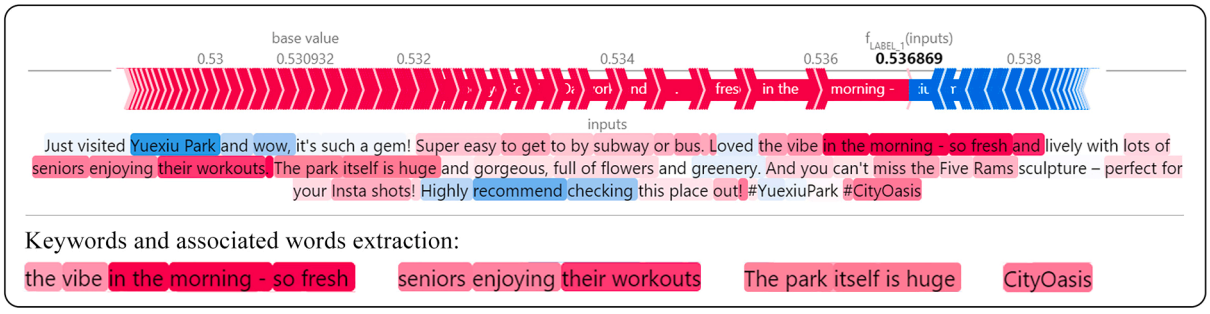
a) Sample park review comment

Just visited Yuexiu Park and wow, it's such a gem! Super easy to get to by subway or bus. Loved the vibe in the morning - so fresh and lively with lots of seniors enjoying their workouts. The park itself is huge and gorgeous, full of flowers and greenery. And you can't miss the Five Rams sculpture – perfect for your Insta shots! Highly recommend checking this place out! #YuexiuPark #CityOasis

b) SHAP-based keywords extraction results of accessibility



c) SHAP-based keywords extraction results of usability



d) SHAP-based keywords extraction results of attractiveness

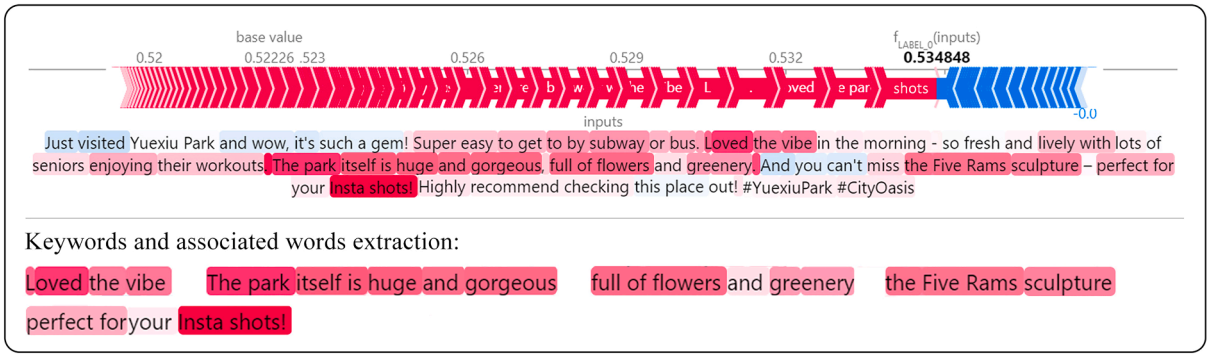


Fig. 7. An example of SHAP-based keyword extraction results.

Table 5
Comparative analysis of keyword extraction.

	Number of comments	Token Count (direct)	Token Count (SHAP-based)	Reduced
Accessibility	13,329	35,972	2933	91.85 %
Usability	24,201	41,102	9842	76.05 %
Attractiveness	46,123	58,006	11,322	80.48 %

the key vocabulary within the comments. This approach offers valuable insights into the PDP model's decision-making process, enhancing both its predictive and interpretive capabilities.

3.3.2. *Results of keywords and clusters extraction*

The optimal number of clusters and the primary themes for each cluster were determined based on the average silhouette score and the interpretability of each cluster (Figs. 8–10). For accessibility, we identified 7 clusters with 44 thematic words. Usability comprised 16 clusters with 68 thematic words, and attractiveness consisted of 12 clusters with 46 thematic words.

For accessibility, the top five thematic words identified were 'subway (entrance/station)', 'public transportation/traffic', 'hour', 'parking lot', and 'walk'. For usability, the top five were 'child/baby', 'running/jogging', 'photography', '(morning) exercise', and 'bicycle/cycling'. In attractiveness, the most frequently mentioned thematic words were 'scenery/landscape', 'environment', 'air', 'feature', and 'check-in

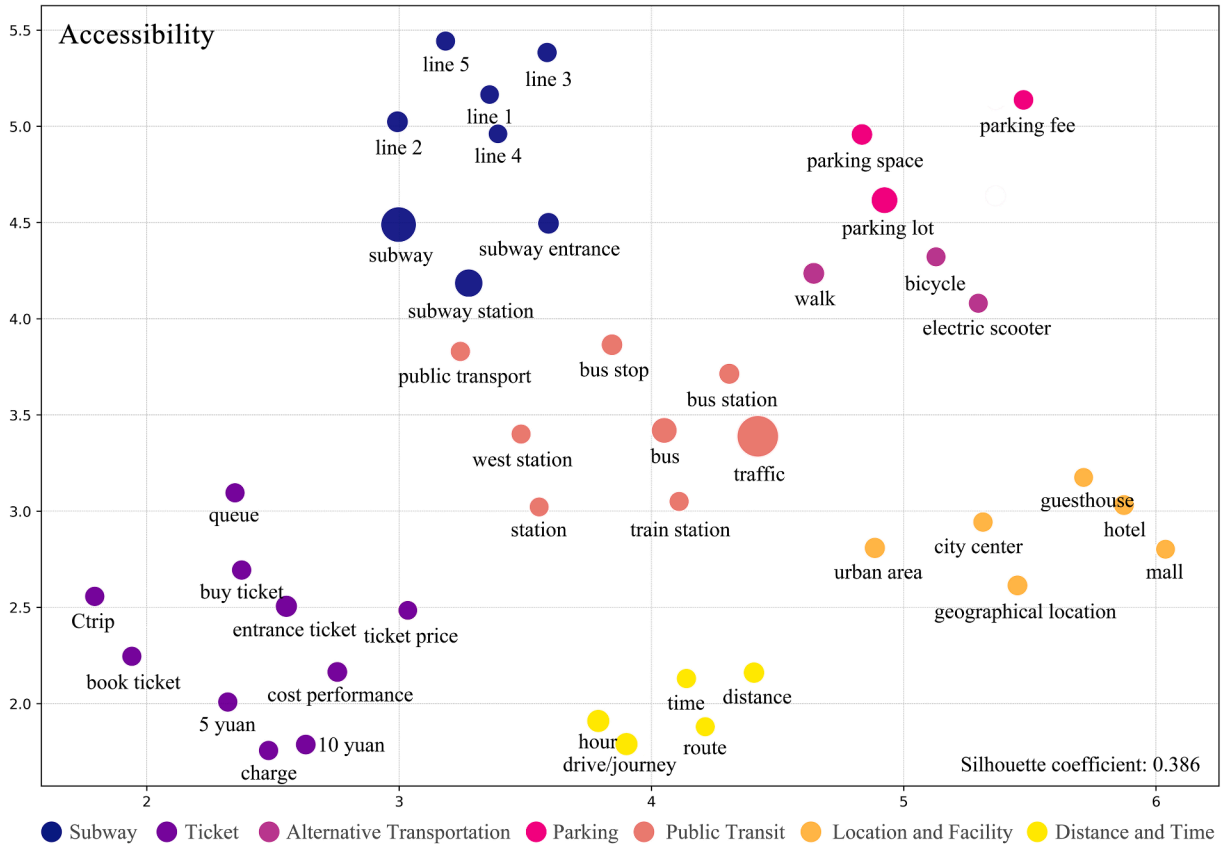


Fig. 8. Clusters related to park accessibility.

photo'.

3.3.3. *Results of associated words extraction*

In our analysis, we observed semantic overlaps among some keywords, such as "line 1", "line 2", and "line 3". These overlapping words were merged during the process of identifying associated words. Ultimately, for accessibility, 23 merged words were identified, yielding 267 effective associated words (Table 6). For usability, 34 merged words led to 375 associated words, and for attractiveness, 26 merged words resulted in 312 associated words (Appendix A).

4. Discussion

4.1. *Methodological contributions*

SMD has been proven to be a valuable data source for quantitative research on landscape perception, value, and preferences due to its low cost, content richness, and voluntariness (Cheng et al., 2021; Huai et al., 2023). Our study took advantage of this resource by using an automated program to collect a massive volume of SMD with a long time span and across a broad geographic distribution. This approach overcomes traditional investigation's usual limitations of small respondent sizes, laborious and costly data collection processes, and confinement to narrow spatial and temporal scales (Zhou et al., 2024).

Current explorations into SMD's potential in assessing park perceptions face several challenges. A primary issue is the lack of datasets curated for specific SMD assessment tasks (Rahman et al., 2023; Tang et al., 2023; Yan et al., 2024), which is essential for assessing perceptions beyond sentiment analysis (He et al., 2023; Huai et al., 2023; Huai & Van de Voorde 2022) Another challenge involves developing methods for integrating text and image analysis to fully assess park perceptions and effectively extract influencing factors, thereby maximizing SMD's full potential to understand these perceptions.

In this study, we developed the Park Dual-modal Perception (PDP) model based on the latest CN-CLIP architecture. This model is capable of simultaneously analyzing texts and images for perception assessment, allowing it to capture a broader context and achieve a deeper understanding of the input data. It was trained on a manually curated dataset, enabling the accurate conversion of unstructured SMD information into quantifiable scores across three dimensions: accessibility, usability, and attractiveness. Impressively, the average accuracy of the PDP model across three dimensions reached 86.81 %, exceeding the performance of models that rely solely on texts or images. This achievement reflects the benefits of integrating both textual and image information into perception prediction tasks. Additionally, it also supports the common practice on social media platforms where users often share texts along with images to express their emotions or viewpoints (Zhu et al., 2022), further demonstrating the superiority of our dual-modal methodology.

Furthermore, developing an effective lexicon is essential for identifying critical themes in social media-based studies (Koblet & Purves, 2020). Based on the PDP model and SHAP, we established a novel method for not just 'assessing' but also 'interpreting' perceptions. The PDP model, which captures comprehensive information from text, employs SHAP for targeted extraction of vocabulary relevant to each dimension. Remarkably, it removed an average of 82.79 % of irrelevant words across all dimensions, efficiently identifying keywords and associated words. This addresses the issue inherent in traditional frequency-based keyword extraction methods, which often struggle with disruptions caused by irrelevant information (Huai & Van de Voorde, 2022; Liu et al., 2023; Tang et al., 2023).

Overall, while our analyses are based on individual comments, our methods effectively integrate these personal insights to extract and summarize general public perceptions. Similar to the work of Zhou et al. (2024), the PDP model's structure allows for application in various

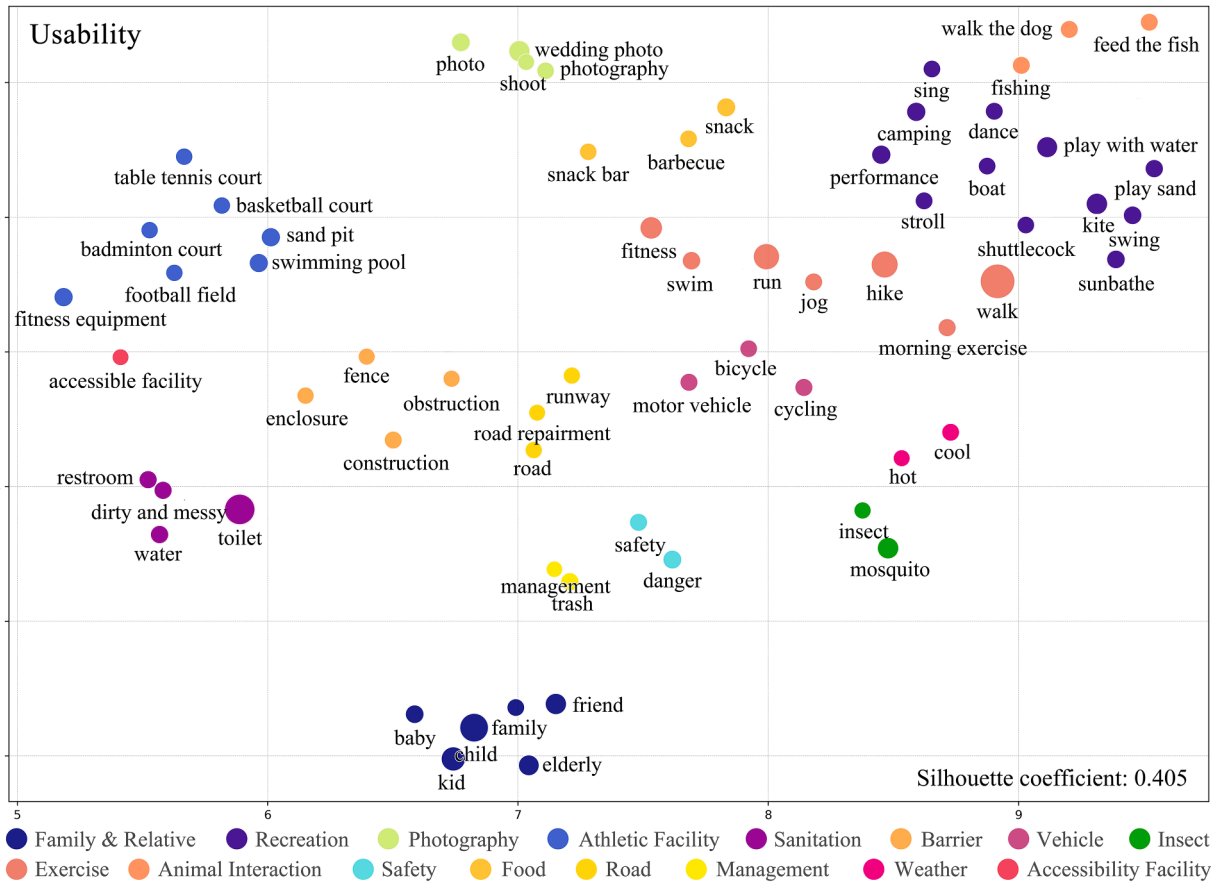


Fig. 9. Clusters related to park usability.

urban contexts. This process involves 1) fine-tuning the text and image encoders on the CLIP model, 2) using the fine-tuned encoders to extract both text and image features, and 3) fusing these features to generate output predictions. To apply the model to new contexts, researchers need to 1) develop a dataset curated for the specific SMD assessment tasks, 2) train the dual-modal model on this dataset, and 3) apply the model to assess and interpret the required tasks. Researchers aiming to adapt this model for different applications can access resources for fine-tuning CLIP at Hugging Face.[1]

4.2. Theoretical implications to park studies

Early machine learning models, though potent in application potential, often face limitations due to their restricted interpretability. Our study, utilizing the SHAP-based keyword extraction method, effectively summarizes specific factors influencing accessibility, usability, and attractiveness of parks from a user experience perspective. This approach not only aligns with but also enriches existing empirical studies derived from traditional surveys, thereby deepening our understanding of park perceptions.

4.2.1. Factors influencing accessibility

For accessibility, travel distance and time are the most fundamental factors affecting park use (Wang et al., 2015). Transportation satisfaction, a frequently mentioned aspect, is intrinsically linked to these factors. This includes public transportation options like subways and buses, with associated words involving “convenient”, “nearby”, “direct accessible”, “available”, and “many”. Moreover, driving to park is a prevalent choice, where “free”, “convenient”, “ample”, and “orderly parking lots” are frequently mentioned. Consistent with previous research, visitors prefer parks with sufficient parking facilities (Li et al., 2023). Insufficient parking or high fees can lead to chaotic parking situations and traffic congestion, negatively impacting the park experience (Bureau, 2023).

Alternative modes of transportation, such as electric scooters, bicycles, and walking are also popular for park visits and exploration. Terms related to walking often highlight proximity, such as "a few minutes" or "a few hundred meters" away. Meanwhile, bicycles and electric vehicles are associated with activities like “riding”, “renting”, “uphill”, “downhill”, and “sightseeing”. People tend to visit parks that are accessible by walking or (electric) bicycles (Li et al., 2021), but high rental costs or bicycle entry restrictions can prompt complaints.

Moreover, the presence of commercial facilities near parks significantly influences visitation (Byrne & Wolch, 2009; Chen et al., 2018; Huai et al., 2023). Hotels, malls, and city centers, as key thematic words from our study, fulfill visitors’ accommodation and shopping needs.

4.2.2. Factors influencing usability

Parks offering various functions and amenities can support a broader range of users (Kaczynski et al., 2008). In our analysis, the most frequently mentioned user groups are “children”, “elderly”, “families”, and “friends”. Positive comments often indicate suitability for “play” and “bringing/taking children”, whereas negative feedback typically relates to “overcrowding” and “not suitable” for the elderly or children. Regardless of age, the availability of facilities forms the foundation for all activities, emphasizing the importance of convenient man-made amenities (Dallimer et al., 2014; Wang et al., 2021). Among these facilities, public sports amenities like “runway”, “table tennis”, and “badminton court” are frequently cited. This underscores the role of

[1] https://huggingface.co/docs/transformers/model_doc/clip

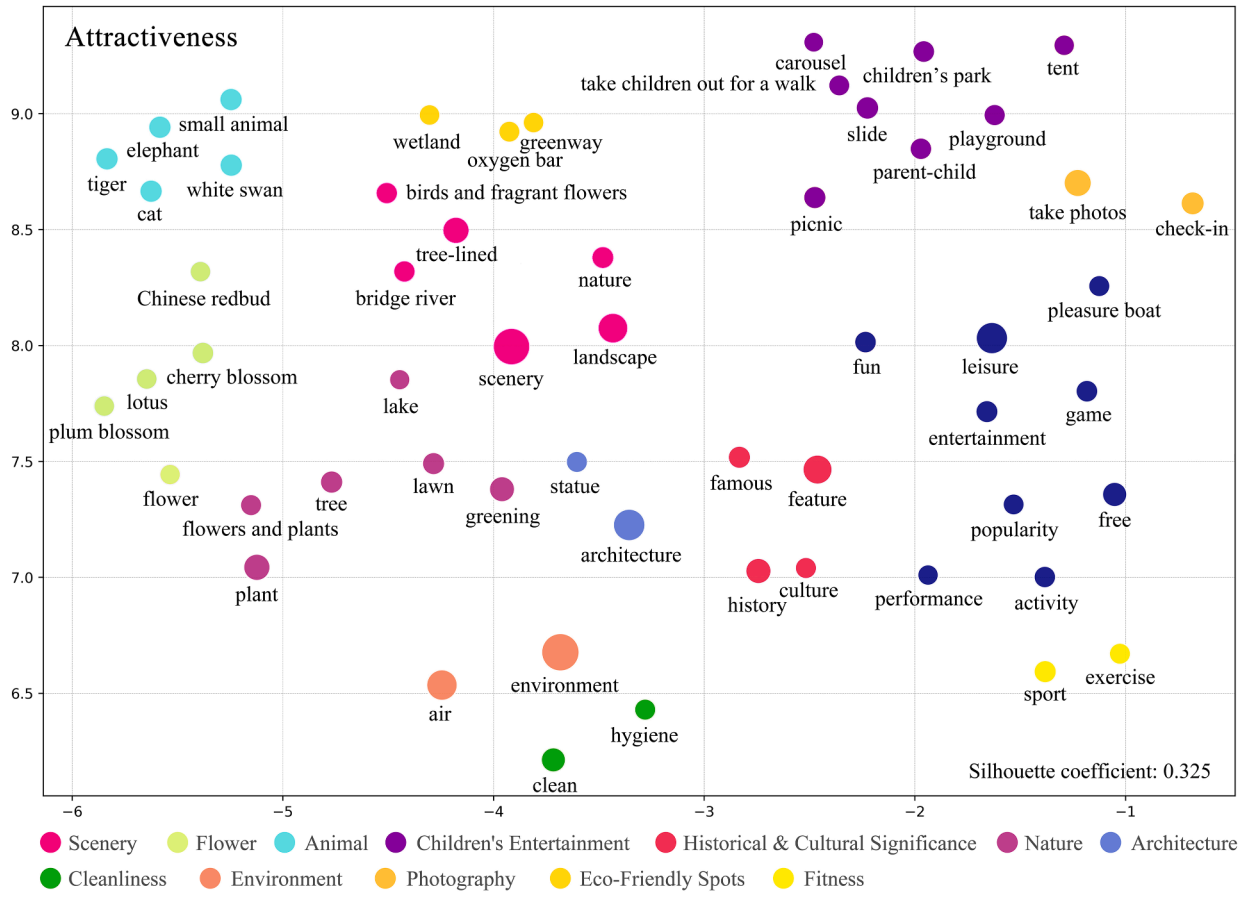


Fig. 10. Clusters related to park attractiveness.

parks as key venues for physical exercise (Kaczynski et al., 2008; Wan et al., 2021; Wolch et al., 2014). Sanitation facilities, including toilets and trash bins, are also a common focus in comments, with well-maintained hygiene playing a significant role in enhancing park usability (Chen et al., 2018; Razak et al., 2016).

The quality of park facilities, encompassing environmental conditions (presence of mosquitoes, safety concerns, perceived danger, heat, and coolness), the capacity to meet user needs, and maintenance aspects (cleanliness, state of repair, construction, and obstructions), are identified as crucial to park usability (Gu et al., 2020; Liu et al., 2023; McCormack et al., 2010; Stessens et al., 2020).

Moreover, people's perception and use of parks depend not only on their cultural backgrounds and values but also on their shared interests in certain activities (Wang et al., 2015). Activities possible within parks have numerous keywords (photography, kite flying, camping, etc.), with positive associated words like "suitable", "available", and "leisure", as well as negative ones such as "unsuitable", "impossible", and "crowded". Organizing activities that align with residents' common interests can significantly enhance park usability.

4.2.3. *Factors influencing attractiveness*

The attractiveness of parks is directly linked to both their cultural (e.g., aesthetic experiences, leisure entertainment, and place perception) and regulatory characteristics (e.g., air quality, quietness, and microclimate) of ecosystem services (Pietrzyk-Kaszyńska et al., 2017; Wang et al., 2021). Visual-driven aesthetic appreciation plays a critical role in park attractiveness (Byrne & Wolch, 2009; Razak et al., 2016; Wan et al., 2021). Natural elements like "scenery", "landscape", "flowers", and "animals", along with man-made elements like "architecture" and "sculpture," hold significant visual appeal. They attract visitors for photography and enrich aesthetic experiences, thereby enhancing overall perceived park quality.

Beyond mere admiration, studies indicate that parks with a larger proportion of green surfaces are more attractive to visitors, offering both health and recreational benefits (Kothencz & Blaschke, 2017). Natural features such as "wetlands," "greenways," "plants," and "lakes" are frequently cited for their role in stress reduction and attention restoration (Panagopoulos et al., 2016; Liu et al., 2023), contributing to enhanced well-being (Wan et al., 2021). Lawns are particularly noted for facilitating various recreational activities such as "picnicking", "camping", and "tenting". However, issues like inaccessibility or poor maintenance of these lawns often result in public complaints (Wan et al., 2021).

Moreover, parks support numerous sports and social activities (Ries et al., 2009). "Playgrounds", "children's parks", "slides", and "carousels" provide active play areas for children, fostering socialization and familiarization opportunities among guardians from diverse backgrounds, thereby encouraging spontaneous park usage (Gill & Simeoni, 1995).

4.3. *Implications for urban planning*

In practical applications, urban planners can firstly use our PDP model to quantify public perception scores across the three dimensions and use the visual scoring maps to identify lower-rated parks at the macro scale. Further, through the incorporation of SHAP within the PDP model, they can filter out irrelevant content and precisely identify factors contributing to score variations, providing strategic suggestions for micro-scale enhancements. Taking Guangzhou as an example:

City Scale Analysis (Macro perspective): The visualized scoring maps of 130 parks (Fig. 6) effectively identify score distributions across the three dimensions. For instance, Yuexiu and Liwan districts,

Table 6
Words associated with accessibility.

Cluster name	Thematic word	Positive associated word								Negative associated word				
Subway	Line 1/2/3/4/5 subway (entrance/station)	convenient	nearby	traffic	available	at/in	bus	direct	take	inconvenient	not available	far	construct	isolated
	12,639	3596	2565	1727	1276	1229	659	588	303	304	214	106	52	20
Ticket	book/buy ticket	free admission	can	free	no need	no charge	yuan	worthy		limited	need	require		
	1274	453	211	125	99	89	51	32		86	71	57		
	ticket price/charge/yuan	free admission	no need	free	ticket	require	no	yuan	worthy	a bit expensive	relatively expensive	not enough	require	quite a few
	1282	453	174	125	116	77	89	51	32	43	43	30	27	22
	Ctrip	book ticket	on	efficient	book in advance	discount	take ticket	activity	cheap	less discount	suffered a loss	slow ticket issuance	incorrect time	
	755	226	209	104	48	44	29	21	15	28	14	9	8	
	queue	no need	people	brief duration	convenient	parking	go inside	minutes	take a seat	too long	require	wait		
	918	328	84	78	46	40	27	20	18	124	121	32		
	cost performance	high	very	good	super high	better	can	worthy	awesome	low	not high	ordinary		
	772	256	165	50	29	28	15	14	10	91	78	36		
Alternative Transportation	electric scooter	take	convenient	uphill	downhill	sightseeing	round trip	offer	shuttle	not available	expensive	can only		
	1736	1211	321	45	32	20	18	12	8	39	15	15		
	bicycle	cycling	rent	can	many	nearby	have fun	children	tandem	unable to enter	high rental fee	too hot		
	516	165	102	87	27	17	24	8	8	45	21	12		
	walk	a few minutes	subway	hundred meters	get off	uphill	walk to	navigation	near	tired	far	uphill	can only	inconvenient
	1922	421	329	263	250	125	174	55	45	87	65	57	28	23
	bus	many	direct	convenient	gate	pass	direct	nearby	transfer	inconvenient	not available	far		
	1480	298	298	293	172	101	65	50	31	99	44	29		
Parking	parking space	available	free	convenient	many	plenty	ample	easy to find	roadside	few	not available	shortage	insufficient	not enough
	1210	279	111	100	92	91	74	44	36	108	78	68	65	64
	parking lot	available	free	convenient	charge	large	sufficient	spacious	outdoor	not available	charge	few parking space	small	private
	3157	1240	542	242	242	171	154	81	63	152	125	69	40	36
	parking fee	cheap	no need	free	convenient	cost-effective	one-off	10 yuan	save	expensive	not cheap			
	177	38	37	14	11	10	8	8	6	26	19			
Public Transit	train station	nearby	next to	not far	opposite to	close to	subway	downtown		occupy	noisy			
	135	58	17	14	14	12	7	7		3	3			
	bus stop/station	available	nearby	convenient	gate	subway	across from	next to	take the bus	far	not available	isolated	only have	few
	1269	483	158	173	117	91	62	62	50	33	23	7	6	4
	public transport/traffic	convenient	comfortable	subway	bus	nearby	downtown	direct	take	inconvenient	isolated	far	no bus	
	8240	3222	1398	1372	bus	205	202	144	133	879	82	78	70	

(*continued on next page*)

Table 6 (*continued*)

Cluster name	Thematic word	Positive associated word								Negative associated word								
Location and Facility	hotel/ guesthouse	nearby	live	next to	across from	close to	not far	scenic spot	convenient	parking available only at adjacent hotel	no	few	far					
	395	131	87	39	33	20	16	11	10	20	15	8	5					
	mall	available	across from	nearby	many	shopping	dining	convenient	around	parking available only at opposite mall								
	255	53	39	31	30	26	26	21	15	14								
	city center/ urban area	at/in	is	convenient	bustling	not far	nearby	rare	weekend	far away	distance	driving time						
	1283	303	229	160	64	55	35	21	15	290	66	45						
	geographical location	superior	convenient	extremely	nice	central area	good	blessed with unique advantages		isolated	far							
	467	127	67	60	52	37	44	12		60	8							
Distance and Time	route/ distance/ drive/journey	minute(s)	not far	hour(s)	downtown	kilometer (s)	nearby	meter(s)	children's park	far	a little	relatively	hour(s)	minute(s)				
	1075	134	105	46	45	44	29	24	23	258	230	80	38	19				
	hour	one	half	two	more than one	one and a half	three	several		one	two	more than one	one and a half	half	several	three		
	3768	1210	297	259	245	208	67	58		831	165	163	125	85	55	39		
	time	available	have	sightseeing	hour(s)	free	plenty of	one day	throughout the day	hour(s)	open	no	not enough	short				
	1813	453	380	241	180	125	117	72	70	47	40	37	30	21				

traditional commercial centers, and densely populated areas of Guangzhou, feature parks with high levels of accessibility, usability, and attractiveness due to well-developed infrastructure and concentrated park distribution. In such densely populated districts, introducing micro-parks (such as community parks and pocket parks) and improving older parks (Viinikka et al., 2023) could increase park accessibility, especially for vulnerable groups (Wang et al., 2015; Wendel et al., 2012). Panyu district, being relatively remote, generally has poorer accessibility but features parks with natural beauty that attract visitors year-round. Notably, parks along the subway lines in Panyu form green clusters serving nearby residents, highlighting the importance of enhancing connections between existing parks (Zhang et al., 2015). However, certain parks in this district with poor transportation access score low, suggesting a priority on improving accessibility before enhancing quality (Li et al., 2021). In other districts, the noticeable gap between few high-scoring and many low-scoring parks calls for urban planners to focus on both underused and popular parks. This approach should consider the diverse needs of different social groups and leverage the unique developmental potential for a more inclusive and comprehensive improvement of the park system (Biernacka et al., 2022).

Individual Park Scale Analysis (Micro perspective): Urban planners can use scoring maps to identify parks with lower scores and then apply the SHAP-based keyword extraction method for a detailed review of public critiques and suggestions. For instance, in the Tianhe district, known for its convenient transportation and well-developed infrastructure, parks like Huolushan Forest Park score high in accessibility but lower in usability and attractiveness. SHAP analysis highlights specific concerns (Fig. 11) such as "too far," "difficult to park," and "rough roads," signaling aspects for improvement. Park managers could enhance quality by improving park management, adding restrooms, increasing resting spaces for hiking, and installing clear signage for better guidance. Addressing perceptions of the park being boring or ordinary could be tackled by adding attractive elements like artistic installations (Wan et al., 2021), creating high-quality natural landscapes (Li et al., 2023), and organizing engaging activities (Wang et al., 2021) to enrich the visitor experience.

To clarify, these suggestions, which reflect the general public's opinions, can serve as references for practical park management. However, relying solely on one type of data is insufficient in practice. Urban planners should integrate these suggestions with other data analyses when developing effective multi-scale governance or design actions.

4.4. Limitations and future work

We attempted to discern differences among the three dimensions through image content analysis. While successful in object identification within the images, we observed a uniformity in image content across dimensions, leading to minimal discernible variation. This might be due to the prevalence of images depicting elements like trees, roads, and skies, with less emphasis on negative elements. Future research should investigate methods for more accurately extracting negative factors from images. Besides, while SHAP effectively identifies keywords the PDP model focuses on, calculating SHAP values is time-intensive. Enhancements in deep learning interpretability algorithms are necessary for efficiency. It's also noticeable that our analysis of the reasons for low ratings focused primarily on the most frequently mentioned negative terms (Fig. 11). It is crucial to identify, categorize, and weigh the various 'detractors' more precisely to ascertain whether they can be rectified during the redesign phase. Future work could enhance our SHAP-based keyword extraction method to facilitate a more accurate diagnosis of the underlying issues, thereby improving park management and redesign efforts.

Additionally, we recommend integrating diverse data sources and methodologies to achieve more comprehensive suggestions for park management. Firstly, while our current methods predominantly capture public perceptions from SMD, park planning must also consider critical

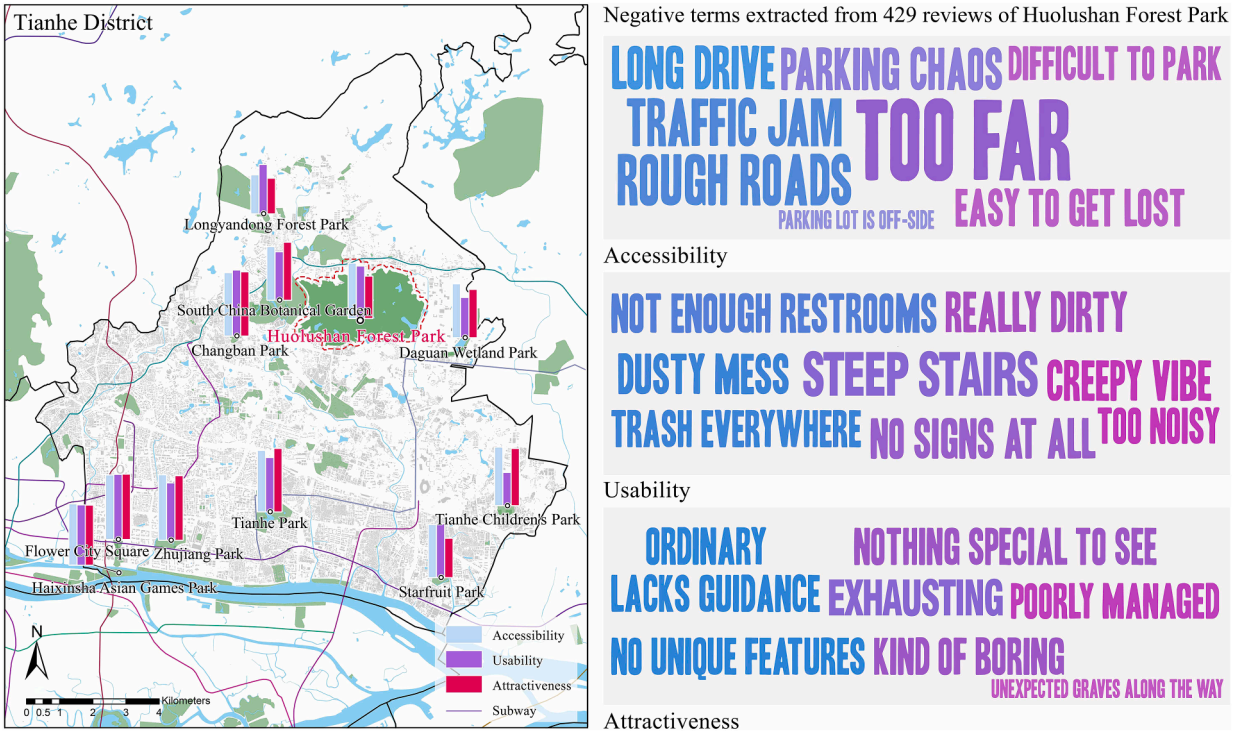


Fig. 11. Negative lexicon in Huolushan Forest Park.

environmental issues such as biodiversity preservation, air quality enhancement, and climate change mitigation—issues that are difficult to assess with SMD alone. Secondly, our ability to extract perceptions and their influencing factors is constrained to the information explicitly mentioned by visitors. When assessing dimensions less frequently mentioned by visitors on social media, the model may struggle to capture the relevant information, which can render the analysis results less reliable. We also acknowledge the challenge in capturing less mentioned socio-demographic information such as gender, age, and cultural backgrounds. Implementing additional research methods, such as distributing targeted questionnaires, could supplement this missing information and ensure that opinions from underrepresented groups are adequately captured. This enriched data would also enable more detailed analyses of perceptions and assist in developing targeted landscape design recommendations that are better tailored to the diverse preferences of various groups. Lastly, future studies should consider how changes in park environments might influence visitors' attitudes and behaviors over time, contributing to a more dynamic understanding of park perception.

5. Conclusion

This study introduces the Park Dual-modal Perception (PDP) model and establishes a workflow that synergizes the assessment and interpretation of park perceptions using the PDP model, thereby advancing the use of Social Media Data (SMD) to better understand public perceptions of urban parks.

Methodologically, our SMD-based approach addresses the common limitations of traditional surveys, which typically involve small respondent sizes, and labor-intensive data collection, and are usually confined to narrow spatial and temporal scales. Moreover, while most park perception studies focus solely on sentiment analysis, our manually annotated park perception dataset explores broader dimensions of perception, setting a precedent for future urban studies. In addition, the PDP model effectively integrates text and image analysis from comments, addressing the common issue in related studies that rely on a single modal of data. This model achieves an average accuracy of 86.81 %, significantly outperforming commonly used models such as BERT, LSTM, and XGBoost.

Utilizing ArcMap to visualize the model's assessment results, we generate scoring maps that identify parks with lower perceived values at an urban scale. Further incorporation of SHAP within the PDP model eliminates 82.79 % of irrelevant words, addressing the limitations of common word frequency analysis, which often struggles with excessive irrelevant information and fails to focus on specific perceptual dimensions. This targeted analysis helps clarify the factors contributing to score variations and supports targeted recommendations for park-level improvements.

On the application front, the adaptable framework of the PDP model can be trained on other datasets, extending its applicability beyond park-centered urban planning to broader urban management practices. This adaptability fosters a comprehensive understanding of public perceptions, enhancing public engagement in urban planning and supporting policymakers and stakeholders in identifying and addressing urgent improvement needs.

Theoretically, our study identifies specific factors influencing perceived park accessibility, usability, and attractiveness from a user-experience viewpoint, pinpointing 158 thematic words and 954 associated words. Key findings indicate that factors such as distance, travel time, transportation mode, ticket prices, and proximity to commercial amenities are pivotal in determining park accessibility. Park usability hinges on its ability to serve diverse user groups and provide well-maintained, multifunctional facilities. Park attractiveness is closely linked with the cultural (e.g., aesthetic experience, leisure and entertainment, place perception) and regulatory characteristics (e.g., air quality, quietness, microclimate) of ecosystem services.

While advocating for the use of SMD to optimize site design informed by public perception, this study also acknowledges its current limitations. Expanding the number of city samples and integrating various data sources are recommended to increase the model's universality and robustness. Enhancing the interpretability of deep learning algorithms remains crucial for more effectively extracting latent information from

images and texts.

CRediT authorship contribution statement

Xukai Zhao: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Yuxing Lu:** Writing – review & editing, Validation, Software, Methodology, Conceptualization. **Wenwen Huang:** Writing – review & editing, Data curation, Conceptualization. **Guangsi Lin:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2024.105619.

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